SHADOW FDP — PAPER TO THE CHAIRMAN

Limbayong Deepwater Development · Board Decision Paper

Prepared by HERMES (Shadow Chairman's Office) · 19 October 2026

For the attention of: Tan Sri Dato' Seri Mohd Bakke Salleh, Chairman, PETRONAS
Classification: Shadow paper. Independent. Not a PETRONAS document. No internal data used.

> *Tan Sri, this paper is written in your language, not the geologists'. It has one page of decision, four pages of evidence, and a list of the things I could not see. Where I have guessed, I say so. Where I do not know, I say that too. — H.*

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1. DECISION REQUIRED

Whether to continue committing irrevocable capital to Limbayong (FID taken 28 August 2026) before three specific facts are established.

Recommendation: HOLD capital release for six months. Gate on 1 April 2027.

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2. THE EQUITY QUESTION — ANSWERED FIRST, BECAUSE IT IS THE REAL QUESTION

You asked what equity assumption sits under this project. The answer is the finding:

The equity is 100% PETRONAS Carigali. There is no partner. There will not be one.

That is not a modelling convenience. It is a structural fact, and it is the most important fact in this paper.

Layer	Value	Status
Working interest	PCSB **100%**, operator	[M] Welligence, Aug 2026
Farm-out partner	**None**	[M] 24 years, four failed development attempts
Funding structure	100% equity from PETRONAS balance sheet; no project finance	[A] — no public debt facility disclosed

Layer	Value	Status
Peak cash exposure	**US$4.05bn by 2030**, before first revenue	[D]
RM equivalent	**RM 16.6 billion** at 4.10	[D]
Cash returned	**Not before 2037** — ten years after first spend	[D]
Downside absorbed by	PETRONAS alone, therefore the Malaysian public	[D]

Why 100% is not a choice

A 100% interest is normally a strategic decision. Here it is a **residual** — the position left over after everyone else declined.

Section 7 of the underlying study ran the numbers from a foreign contractor's seat. On a contractor basis, Limbayong needs **US$96/bbl Brent to return 10%**. Not 15% — 10%. Brent has exceeded $96 for a minority of the last decade. That is why the field was discovered in 2002, had a subsea contract awarded in 2021, watched its FPSO tender collapse twice, was declared "commercially unviable as a standalone" in 2023, and only reached FID in 2026 — when a used hull was bought for US$35 million.

I stress-tested whether a farm-out could fix this. For a 49% partner to reach a 15% IRR at $70 Brent, the fiscal terms required are beyond anything Malaysia publishes. The project does not generate enough value at $70 to pay a partner a commercial return and leave PETRONAS whole. **Any farm-out would be a transfer of value from the nation to a private party, dressed as risk transfer.** If a farm-out proposal reaches your board, that is the question to ask it.

What this means for the equity assumption

Every model in this pack — mine and the eight technical sections — runs at **100% equity, no debt, 10% real discount rate**. That is correct for PCSB. But it carries a consequence that a chairman should see clearly:

There is no risk-sharing layer between this project and the national balance sheet.

No partner to share the overrun. No lender to test the assumptions. No joint-venture committee to ask the awkward question. When a 100%-owned project is wrong, it is wrong in private, and nobody outside finds out until the write-down.

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3. THE THREE NUMBERS

Metric	Value at US$70 Brent	At today's US$105
NPV10	**−US$668M**	+US$586M
IRR	**6.0%**	13.3%
Brent required for NPV = 0	**US$88/bbl**	—

Metric	Value at US$70 Brent	At today's US$105
Brent required for 15% IRR	**US$115/bbl**	—

Read the last two lines together. The project breaks even at $88 and needs $115 for a proper return. Brent is $105 today, held there by a Middle East risk premium, not by supply scarcity. **This is not an oil project with a price risk. It is a price bet with an oil project attached.**

Change one variable — capex — and the picture moves more than any oil price move:

Capex	capex/boe	NPV10 @ $70	IRR	Breakeven Brent
US$2.90bn	$10.5	−US$180M	8.6%	$68
US$4.05bn (WBS P50)	**$15.0**	**−US$668M**	**6.0%**	**$88**
US$6.07bn (P50 +50%)	$22.0	−US$2,281M	−0.2%	$133

US$15/boe is the industry-typical deepwater number. US$10.5/boe is not. **The estimate that makes this project look acceptable is the one below the global range.** That should worry the board more than the oil price.

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4. WHAT IS ACTUALLY KNOWN, AND WHAT IS NOT

Item	Status	Consequence
Reserves 139 MMbbl + 791 bcf	**Known** — public tracker, data year 2023	The only hard number in the pack
Reservoir quality	**Known** — operator's own SPE paper: *"industry top decile reservoir complexity"*, *"lateral compartmentalization"*	Sanctioning basis, in PETRONAS's own words
Appraisal coverage	**Known** — 4 wells across 30 km of strike, one per 7.5 km	10 development wells are de facto appraisal
Well cost	**Estimated** — US$67M/well P50, no Malaysian deepwater benchmark published	±40%
Capex total	**Estimated** — US$4.05bn P50, AACE Class 4	±30/+50%
FPSO conversion cost	**NOT KNOWN** — the only public figure I found was a 2011 article, withdrawn	**Critical gap**
FPSO capacity fit	**Known gap** — Excalibur ~25–30 kbopd vs 40 kbopd design; no gas handling	31% short on oil, 100% short on gas
First oil date	**Estimated** — Q4 2030 P50; comparables took 43–46 months on clean hulls	2029 is not credible

Item	Status	Consequence
Gas sales agreement	**NOT KNOWN** — no public GSA, no defined route	US$317M of the NPV
PSC fiscal vintage	**NOT KNOWN** — deepwater R/C terms are bespoke and unpublished	±8 months of payback

Two of these gaps are disqualifying at board level. A chairman cannot approve US$4.05bn of capital when the conversion cost of the central facility is unknown and the gas — 39% of the barrels and 110% of the NPV — has no buyer.

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5. WHAT WOULD MAKE ME CHANGE MY MIND

Five conditions. Three unknowns, two failures.

#	Condition	Status	Evidence needed	Cost	Time
1	Excalibur, after conversion, delivers 40 kbopd + 150 mmscfd	**UNKNOWN**	Lump-sum conversion scope from MMHE; class survey of hull and mooring	US$8–12M	4–5 mo
2	Real capex ≤ US$2.9bn	**FAILS** on WBS	Independent cost assurance against P50	US$2–3M	3 mo
3	10 wells deliver ≥35 kboe/d; P90 ≥120 MMbbl	**UNKNOWN**	One appraisal well on lateral continuity + DST	US$93–125M	9–14 mo
4	Gas sold ≥US$6/mmbtu, route defined	**UNKNOWN**	Gas sales agreement; SSGP Kimanis tie decision	US$40–60M	6 mo
5	Brent ≥US$88 sustained to 2030	**Met today — not controllable**	Nothing. This is the bet.	—	—

Score: 0 TRUE · 3 UNKNOWN · 2 FALSE.

A total of ~US$60M and six months converts three of these into answers. That is 1.5% of the capital at risk, to remove the possibility of a US$4bn mistake. **The asymmetry is not close.**

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6. THE ARGUMENT FOR PROCEEDING ANYWAY — AND ITS HONEST PRICE

I would be a poor shadow chairman if I only gave you the numbers that support a HOLD. There is a real case for proceeding, and it is not the one in the press release.

The strategic case: PCSB is 100% WI, it is the operator, and it has never led a Malaysian deepwater development. A host at Limbayong is the only infrastructure that monetises Bestari and anything else found along the L-B-P trend. A host for one marginal field is a stranded asset. A host with a second customer is a business. Judged on that basis, Limbayong is not a field development — it is the **entry ticket to Sabah deepwater as an operator.**

The price of that argument: it is a portfolio and capability decision, and it should be approved as one. It should not be approved on the oil economics, because the oil economics do not support it at any price below $88.

A sovereign operator is entitled to make strategic investments. It is not entitled to describe them as commercial ones. **If the board approves Limbayong, approve it as a capability investment with a stated strategic return, not as an NPV-positive project. Then the accountability is honest, and the next chairman inherits a decision that can be evaluated against what it claimed.**

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7. WHAT I COULD NOT SEE

Stated plainly, because a paper that hides its blindness is worse than no paper.

- No PETRONAS reservoir model, well proposal, AFE, FDP, or gas contract.
- No MMHE conversion estimate. **The number I most needed, I could not find.**
- No class survey or condition report on Excalibur after six years laid up.
- No PSC text. Deepwater R/C terms are bespoke and not published; I modelled a generic cost-recovery structure with a profitability sliding scale.
- No well performance data from any Malaysian deepwater analogue.
- Every subsurface number is analogue-derived. There is no public seismic, log, core or PVT for this field, and I have not invented one.

A full FDP would cost **US$15–40M and take 12–18 months** of institutional work. This paper cost a fraction of that and answers a different question: **not "what is the answer" but "what would we need to know before the answer can be trusted."**

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8. RECOMMENDATION

HOLD. Six months. Three deliverables. One gate.

By	Deliverable	Owner	Cost
Feb 2027	Independent cost assurance vs the US$4.05bn WBS	Group Finance + external	US$2–3M
Mar 2027	Lump-sum conversion scope at MMHE, incl. gas handling and 40 kbopd topsides	MISC + PCSB Facilities	US$8–12M
Mar 2027	Gas sales agreement, or a written decision that the gas is stranded	Group Commercial	US$40–60M
1 Apr 2027	**Gate 1: GO / NO-GO / REVIEW**	**Board**	—

If all three land favourably: proceed with a full FDP and a re-baselined schedule. If any fails: the project is a hull write-down of US$35M plus sunk engineering — recoverable, and small against what an unbounded commitment would cost.

The asymmetry is the whole argument. Six months and US$60M, to avoid the possibility of committing US$4.05bn to a project whose central facility cost is unknown, whose gas has no buyer, and whose break-even sits US$17 below today’s oil price.

Respectfully submitted. Every number carries an evidence class. Every claim can be traced. Nothing here is asserted that cannot be checked.
